

Translated

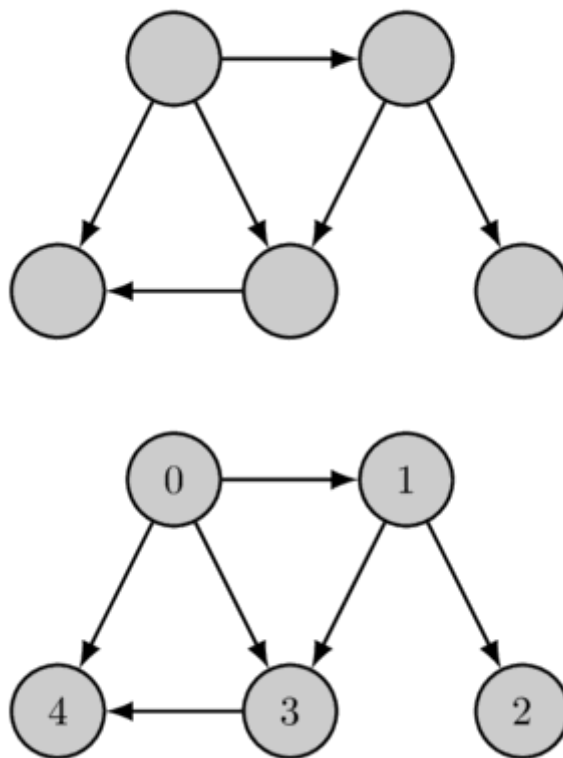
From: e-maxx.ru

Topological Sorting

You are given a directed graph with n vertices and m edges. You have to find an **order of the vertices**, so that every edge leads from the vertex with a smaller index to a vertex with a larger one.

In other words, you want to find a permutation of the vertices (**topological order**) which corresponds to the order defined by all edges of the graph.

Here is one given graph together with its topological order:



Topological order can be **non-unique** (for example, if there exist three vertices a , b , c for which there exist paths from a to b and from a to c but not paths from b to c or from c to b). The example graph also has multiple topological orders, a second topological order is the following: